5-year integrated M.Sc. (Computer science)
Semester-V
Java Programming
Practical Assignment-I

1. Even/Odd Number
Definition: A program to check whether a given number is even or odd.
Explanation: If a number is divisible by 2, it is an even number; otherwise, it is an odd number.

2. Positive, Negative or Zero
Definition: A program to determine whether a given number is positive, negative, or zero.
Explanation: A number greater than 0 is positive, less than 0 is negative, and equal to 0 is zero.

3. Prime Number
Definition: A program to check whether a given number is prime.
Explanation: A prime number has exactly two factors: 1 and the number itself.

4. Palindrome Number
Definition: A program to check whether a given number is a palindrome.
Explanation: A palindrome number remains the same when its digits are reversed.

5. Reverse Number
Definition: A program to reverse the digits of a given number.
Explanation: The program extracts each digit one by one and forms the number in reverse order.

6. Factorial
Definition: A program to calculate the factorial of a given number.
Explanation: The factorial of a number is the product of all positive integers from 1 to that number.

7. Fibonacci Series
Definition: A program to generate the Fibonacci series up to a specified number of terms.
Explanation: In the Fibonacci series, each number is the sum of the two preceding numbers.

8. Sum of Digits
Definition: A program to find the sum of all digits of a given number
Explanation: The program separates each digit and adds them together.

9. Count Digits
Definition: A program to count the number of digits in a given number.
Explanation: The program repeatedly removes the last digit until the number becomes zero, counting each removal.

10. Armstrong Number
Definition: A program to check whether a given number is an Armstrong number.
Explanation: An Armstrong number is a number whose sum of each digit raised to the power of the total number of digits equals the original number.

11. Strong Number
Definition: A program to check whether a given number is a Strong number.
Explanation: A Strong number is a number whose sum of the factorials of its digits equals the original number.

12. Perfect Number
Definition: A program to check whether a given number is a Perfect number.
Explanation: A Perfect number is a number whose proper divisors add up exactly to the number itself.

13. Neon Number
Definition: A program to check whether a given number is a Neon number.
Explanation: A Neon number is a number whose square's digits add up to the original number.

14. Automorphic Number
Definition: A program to check whether a given number is an Automorphic number.
Explanation: An Automorphic number is a number whose square ends with the same digits as the original number.

15. Spy Number
Definition: A program to check whether a given number is a Spy number.
Explanation: A Spy number is a number whose sum of digits is equal to the product of its digits.

16. Leap Year
Definition: A program to determine whether a given year is a leap year.
Explanation: A leap year is divisible by 4, but years divisible by 100 must also be divisible by 400.

17. GCD (Greatest Common Divisor)
Definition: A program to find the Greatest Common Divisor (GCD) of two numbers.
Explanation: The GCD is the largest positive integer that divides both numbers exactly.

18. LCM (Least Common Multiple)
Definition: A program to find the Least Common Multiple (LCM) of two numbers.
Explanation: The LCM is the smallest positive number that is exactly divisible by both given numbers.

19. Power of a Number
Definition: A program to calculate the power of a number.
Explanation: The program multiplies the base by itself repeatedly according to the exponent.

20. Multiplication Table
Definition: A program to print the multiplication table of a given number.
Explanation: The program multiplies the given number by integers in sequence and displays the results.

21. Factors of a Number
Definition: A program to print all factors of a given number.
Explanation: Factors are numbers that divide the given number exactly without leaving a remainder.

22. Prime Numbers in a Range
Definition: A program to print all prime numbers between two given numbers.
Explanation: The program checks each number in the specified range and prints only the prime numbers.

23. Decimal to Binary
Definition: A program to convert a decimal number into its binary equivalent.
Explanation: The program repeatedly divides the decimal number by 2 and stores the remainders to form the binary number.

24. Binary to Decimal
Definition: A program to convert a binary number into its decimal equivalent.
Explanation: The program calculates the decimal value by multiplying each binary digit with the corresponding power of 2.

25. Decimal to Octal
Definition: A program to convert a decimal number into its octal equivalent.
Explanation: The program repeatedly divides the decimal number by 8 and stores the remainders.

26. Decimal to Hexadecimal
Definition: A program to convert a decimal number into its hexadecimal equivalent.
Explanation: The program converts the decimal number into base-16 using repeated division.

27. Binary to Octal
Definition: A program to convert a binary number into its octal equivalent.
Explanation: The program converts binary data into octal format using grouping or decimal conversion.

28. Binary to Hexadecimal
Definition: A program to convert a binary number into its hexadecimal equivalent.
Explanation: The program converts binary digits into hexadecimal representation.

29. Decimal to Roman Number

Definition: A program to convert a decimal number into its Roman numeral equivalent.

Explanation: The program converts integers into Roman numerals using predefined symbols.

30. Happy Number

Definition: A program to check whether a given number is a Happy number.

Explanation: A Happy number eventually becomes 1 when repeatedly replacing the number with the sum of the squares of its digits.

31. Disarium Number

Definition: A program to check whether a given number is a Disarium number.

Explanation: A Disarium number is a number whose digits raised to their respective positions sum to the original number.

32. Duck Number

Definition: A program to check whether a given number is a Duck number.

Explanation: A Duck number contains at least one zero but does not begin with zero.

33. Sunny Number

Definition: A program to check whether a given number is a Sunny number.

Explanation: A Sunny number is a number whose next number is a perfect square.

34. Peterson Number

Definition: A program to check whether a given number is a Peterson number.

Explanation: A Peterson number is a number whose sum of the factorials of its digits equals the original number.

35. Krishnamurthy Number

Definition: A program to check whether a given number is a Krishnamurthy number.

Explanation: A Krishnamurthy number is a special number whose digits' factorial sum equals the number itself.

36. Buzz Number

Definition: A program to check whether a given number is a Buzz number.

Explanation: A Buzz number ends with 7 or is divisible by 7.

37. Magic Number

Definition: A program to check whether a given number is a Magic number.

Explanation: A Magic number becomes 1 after repeatedly summing its digits until a single digit remains.

38. Emirp Number

Definition: A program to check whether a given number is an Emirp number.

Explanation: An Emirp number is a prime number whose reverse is also a different prime number.

39. Circular Prime

Definition: A program to check whether a given number is a Circular Prime.

Explanation: A Circular Prime remains prime after every rotation of its digits.

40. Twin Prime Numbers

Definition: A program to check whether two given prime numbers differ by 2.

Explanation: Twin primes are two prime numbers with a difference of exactly two.

41. Strong Prime Number

Definition: A program to check whether a prime number is greater than the average of its neighboring prime numbers.

Explanation: The program compares a prime number with the average of the previous and next prime numbers.

42. Perfect Square

Definition: A program to determine whether a number is a perfect square.

Explanation: A perfect square is obtained by squaring an integer.

43. Perfect Cube

Definition: A program to determine whether a number is a perfect cube.

Explanation: A perfect cube is obtained by cubing an integer.

44. Sum of Prime Numbers

Definition: A program to calculate the sum of all prime numbers within a given range.

Explanation: The program checks each number for primality and adds only the prime numbers.

45. Prime Factors

Definition: A program to find all prime factors of a given number.

Explanation: The program identifies factors that are themselves prime numbers.

46. HCF and LCM of Three Numbers

Definition: A program to calculate the HCF and LCM of three given numbers.

Explanation: The program extends the HCF and LCM calculation to three integers.

47. Number Frequency

Definition: A program to count the frequency of each digit in a given number.
Explanation: The program counts how many times each digit appears in the number.

48. Decimal to Words
Definition: A program to display a given number in words.
Explanation: The program converts each digit of the number into its corresponding word.

49. Pascal Triangle
Definition**:** A program to print Pascal's Triangle.
Explanation: Pascal's Triangle is generated using combinations or the sum of adjacent elements from the previous row.

50. Floyd's Triangle
Definition: A program to print Floyd's Triangle.
Explanation: Floyd's Triangle displays consecutive natural numbers in a triangular pattern.